

Supervised Learning - Graded Assignment

Predicting Employee Attrition using Supervised Learning Background

Dataset used: IBM HR Analytics Employee Attrition Dataset
(<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>)

Task 1: Data Understanding and Exploration

Dataset Dimensions: (1470, 35)

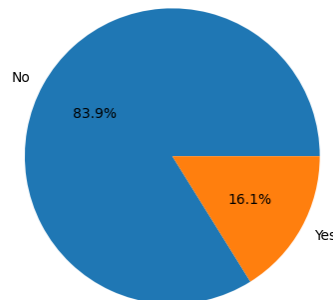
Feature types:

The IBM HR Analytics Employee Attrition dataset contains **35 features**, comprising **26 numerical** (`int64`) **features** and **9 categorical** (`object`) **features**. The numerical features represent measurable employee attributes such as age, monthly income, total working years, and job level, while the categorical features include attributes such as attrition status, business travel, department, education field, gender, job role, marital status, overtime, and over-18 status. This combination of numerical and categorical variables provides a comprehensive representation of employee demographics, work characteristics, and organizational factors for predicting employee attrition.

Target Class Distribution: The target variable in this dataset is 'Attrition'.

Attrition	Count	Percentage
NO	1233	83.877551
YES	237	16.122449

Target Class Distribution - Attrition



Missing Values:

The dataset was examined for missing values using the `isnull().sum()` function. The analysis revealed that there are **no missing values** in any of the features. Therefore, no missing value treatment or imputation was required before model development.

Summary Statistics:

Summary statistics were generated for the numerical features in the dataset. The statistics include the count, mean, standard deviation, minimum, quartiles (25%, 50%, and 75%), and maximum values for each feature. These measures provide an overview of the distribution, central tendency, and variability of the data, helping to identify the range and spread of employee-related attributes before model development.

Statistic	Age	MonthlyIncome	DailyRate	TotalWorkingYears	YearsAtCompany
Count	1470	1470	1470	1470	1470
Mean	36.923810	6502.93	802.49	11.28	7.01
Std	9.135373	4707.96	403.51	7.78	6.13
Min	18.0000	1009.00	102.00	0	0
25%	30.0000	2911.00	465.00	6.00	3.00
50%	36.0000	4919.00	802.00	10.00	5.00
75%	43.0000	8379.00	1157.00	15.00	9.00
Max	60.0000	19999.00	1499.00	40.00	40.00

Three Meaningful Observations:

1. Class Imbalance: The target Variable (Attrition) is imbalance, with significantly more employees staying in the organisation than leaving. This indicates that employee attrition is relatively uncommon, making it important to consider evaluation metrics beyond accuracy, such as precision, recall, F1-score and ROC-AUC.
2. No missing Values: The dataset contains 1470 employee records and 35 features with no missing values. This eliminates the need for missing value imputation and ensures that all records can be used for model training.
3. Wide variation in employee characteristics: The summary stats reveal considerable variation in employee attributes. For example, employees range in age from 18 to 60 years, total working experience ranges from 0 to 40 years, and years at the company range from 0 to 40 years, indicating a diverse workforce with varying levels of experience.

Task 2: Data Preprocessing

Step1: Encoding the target variable

Since the target variable 'Attrition' contains categorical values, it was converted into numerical values (1 for Yes, 0 for No). Machine learning classification algorithms require the target variable to be in numerical form to perform training and prediction.

Step 2: One-Hot Encoding of Categorical Features

The remaining categorical features such as BusinessTravel, Department, EducationField, gender, JobRole, MaritalStatus, OverTime and Over18 were transformed using One-Hot Encoding. This converts each category into separate binary (0 or 1) columns, allowing machine learning algorithms to process categorical information without introducing an artificial order between categories.

Step 3: Feature Scaling

The numerical features were standardized using StandardScaler. Feature scaling ensures that all numerical variables have a similar scale, preventing features with larger numerical values such as MonthlyIncome, from dominating those with smaller values such as Age. Standardization improves the performance of distance-based and gradient-based machine learning algorithms.

Step4: Train-Test Split

The preprocessed dataset was divided into training (80%) and testing (20%) sets using the `train_test_split()` function. The training set was used to train the machine learning models, while the testing set was reserved for evaluating their performance on unseen data. Stratified sampling was applied to preserve the original class distribution of the target variable in both subsets.

Task 3: Model Development

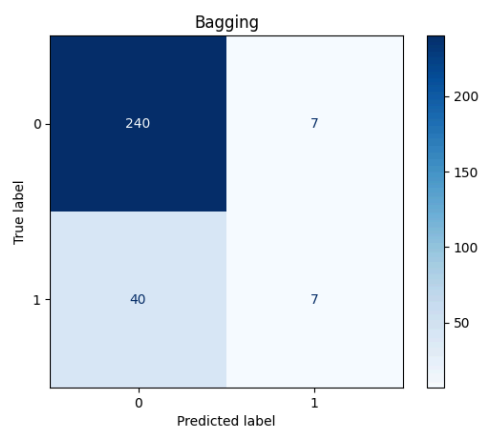
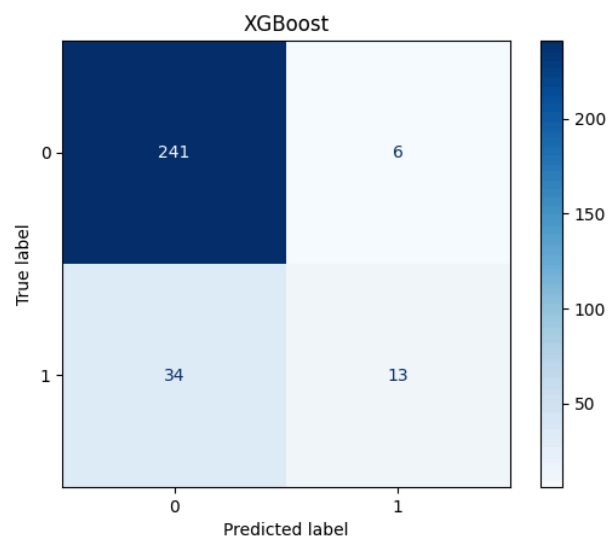
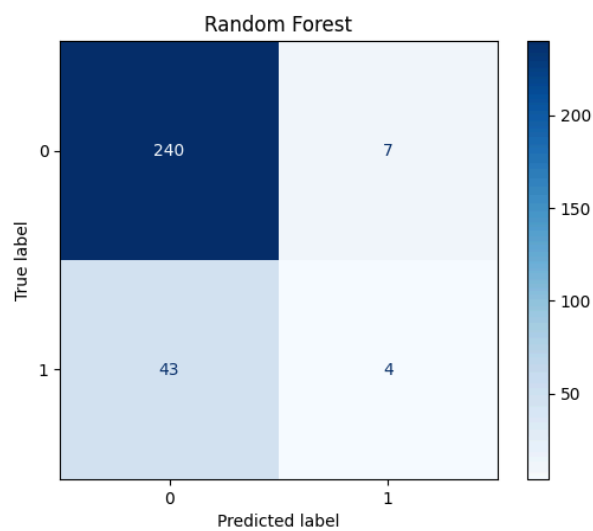
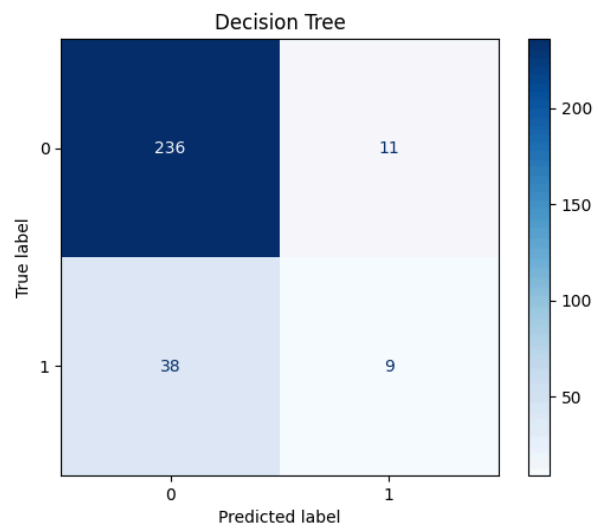
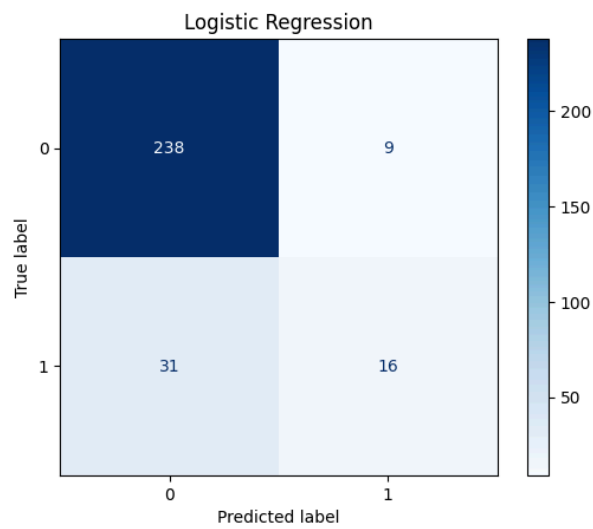
Five classification algorithms were implemented to predict employee attrition. Each model was trained using the training dataset and evaluated on the testing dataset. The selected models represent a combination of linear, tree-based, ensemble, and boosting techniques, allowing their performance to be compared on the same dataset.

Model	Hyperparameters
Logistic Regression	random_state=42 , max_iter=1000
Decision tree	criterion='gini' , max_depth=5 , random_state=42
Random forest	n_estimators=100 , random_state=42
Bagging Classifier	n_estimators=100 , random_state=42
XGBoost	n_estimators=100 , learning_rate=0.1 , random_state=42

Task 4: Model Evaluation

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.863946	0.640000	0.340426	0.444444	0.651994
Decision tree	0.833333	0.450000	0.191489	0.268657	0.573477
Random Forest	0.829932	0.363636	0.085106	0.137931	0.528383
XGBoost	0.863946	0.684211	0.276596	0.393939	0.626152
Bagging	0.840136	0.500000	0.148936	0.229508	0.560298

Confusion Matrices



Strengths of each model:

Model	Strengths
Logistic Regression	Simple, interpretable, fast to train, suitable for binary classification.
Decision Tree	Easy to understand and visualize, captures non-linear relationships.
Random Forest	High accuracy, reduces overfitting, robust to noisy data.
Bagging Classifier	Improves stability and reduces variance by combining multiple models.
XGBoost	Excellent predictive performance, handles complex relationships, highly efficient.

Weaknesses of each model:

Model	Weaknesses
Logistic Regression	Assumes a linear relationship between features and the target.
Decision Tree	Can overfit the training data if not properly controlled.
Random Forest	Less interpretable and requires more computational resources.
Bagging Classifier	More computationally expensive than a single decision tree.
XGBoost	Requires careful hyperparameter tuning and takes longer to train.

Where one model may be preferred:

- **Logistic Regression** - is preferred when model interpretability and fast training are important.
- **Decision Tree** is useful when simple decision rules and visualization are required.
- **Random Forest** is suitable for achieving high accuracy while minimizing overfitting.
- **Bagging Classifier** is beneficial when improving the stability of decision tree models.
- **XGBoost** is preferred for maximizing predictive performance, especially on complex datasets.

Task 5: Model Selection and Reflection

a). Which model would you recommend to the HR Department?

Based on the evaluation results, **Logistic Regression** is recommended for predicting employee attrition. Although Logistic Regression and XGBoost achieved the same accuracy (86.39%), Logistic Regression obtained the **highest Recall (0.3404)**, **highest F1-score (0.4444)**, and **highest ROC-AUC (0.6520)** among all the evaluated models. These results indicate that Logistic Regression provides the best overall balance between correctly identifying employees who are likely to leave and minimizing incorrect predictions.

b). Justify your choice

Precision

Logistic Regression achieved a **Precision of 0.64**, indicating that 64% of the employees predicted to leave actually left the organization. This helps reduce unnecessary HR interventions for employees who are unlikely to resign.

Recall

The model achieved the **highest Recall (0.3404)** among all the evaluated models. Since employee attrition prediction aims to identify employees who are at risk of leaving, recall is an important metric because missing an at-risk employee may result in the loss of valuable talent.

F1-score

Logistic Regression achieved the **highest F1-score (0.4444)**, demonstrating the best balance between precision and recall. This makes it well suited for the employee attrition dataset, where the target classes are imbalanced.

ROC-AUC

The model achieved the **highest ROC-AUC score (0.6520)**, indicating better discrimination between employees who are likely to leave and those who are likely to stay compared to the other models.

Interpretability

One of the major advantages of Logistic Regression is its high interpretability. The relationship between the input features and employee attrition can be understood through the model coefficients, making it easier for HR professionals to explain and trust the predictions.

Generalisation Ability

The model demonstrated good generalisation ability by maintaining strong performance on the unseen testing dataset. Logistic Regression is less prone to overfitting than more complex ensemble models, making it a reliable choice for practical applications.

Practical Deployment Considerations

Logistic Regression is computationally efficient, requires less training time, and is easy to deploy and maintain. These advantages make it a practical solution for HR departments that require a fast, reliable, and interpretable employee attrition prediction model.

c). Discuss whether any of your models exhibit signs of overfitting or underfitting

Based on the evaluation results, the **Decision Tree, Random Forest, Bagging Classifier, and XGBoost** achieved lower Recall, F1-score, and ROC-AUC scores compared to Logistic Regression, suggesting that they did not generalize as effectively on the testing dataset. While the training performance was not compared directly, Decision Trees are generally more susceptible to overfitting due to their tendency to learn complex decision boundaries. Logistic Regression, on the other hand, demonstrated more consistent performance across the evaluation metrics and showed no obvious signs of poor generalisation on the testing data.

d). Suggest two possible improvements that could further improve model performance if additional time or resources were available.

1. Hyperparameter Tuning

The performance of the classification models could be improved by optimizing hyperparameters using techniques such as **Grid Search** or **Randomized Search** with cross-validation. This would help identify the optimal parameter settings for each algorithm.

2. Handling Class Imbalance

Since the employee attrition dataset is imbalanced, techniques such as **SMOTE (Synthetic Minority Oversampling Technique)** or assigning **class weights** could be applied to improve the model's ability to correctly identify employees who are likely to leave, thereby increasing recall and F1-score.
